TOPIC: HEALTH

The topic I took for this project is Health, especially in Australia. Then after I explored the data on the Our World in Data web, I was interested in taking 3 datasets:

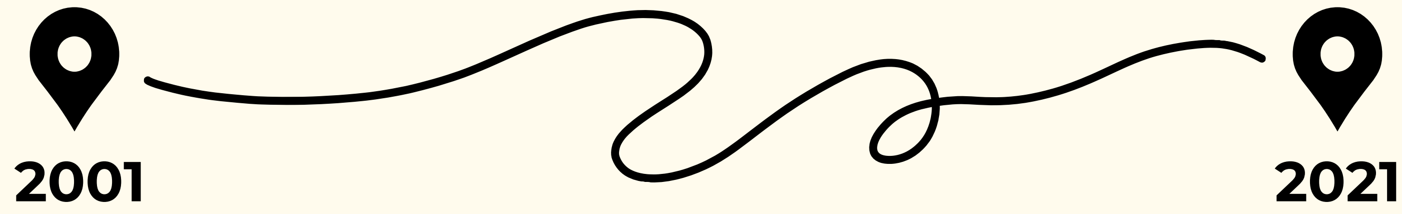
- Population growth
- Healthcare Spending
- Causes of Death

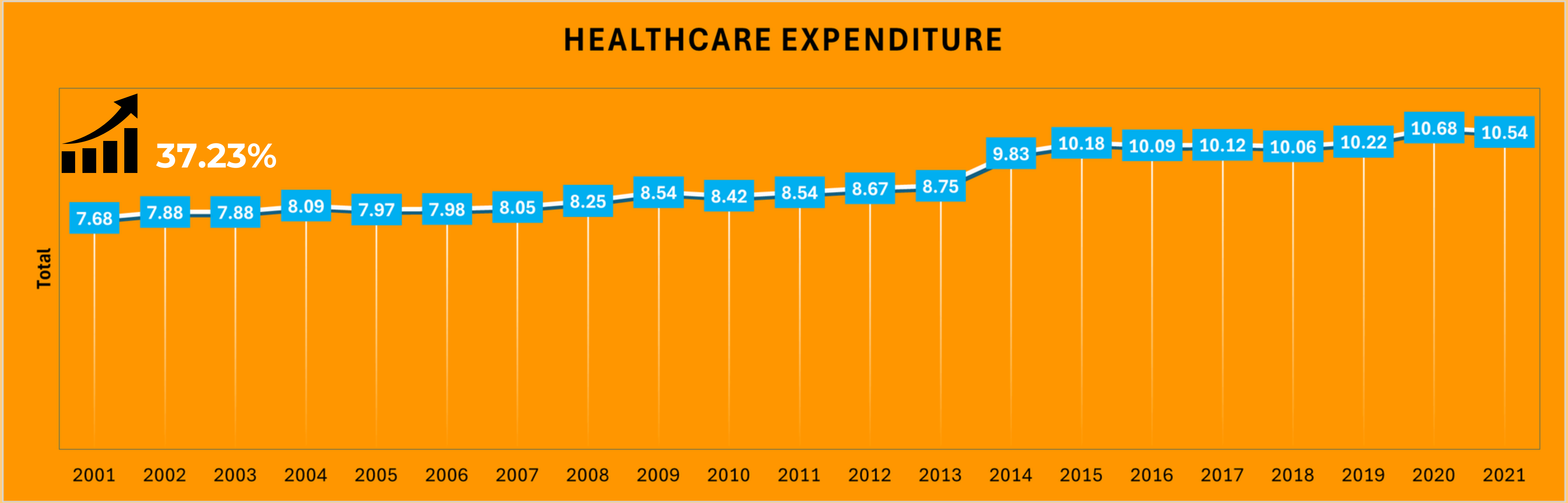
QUESTIONS:

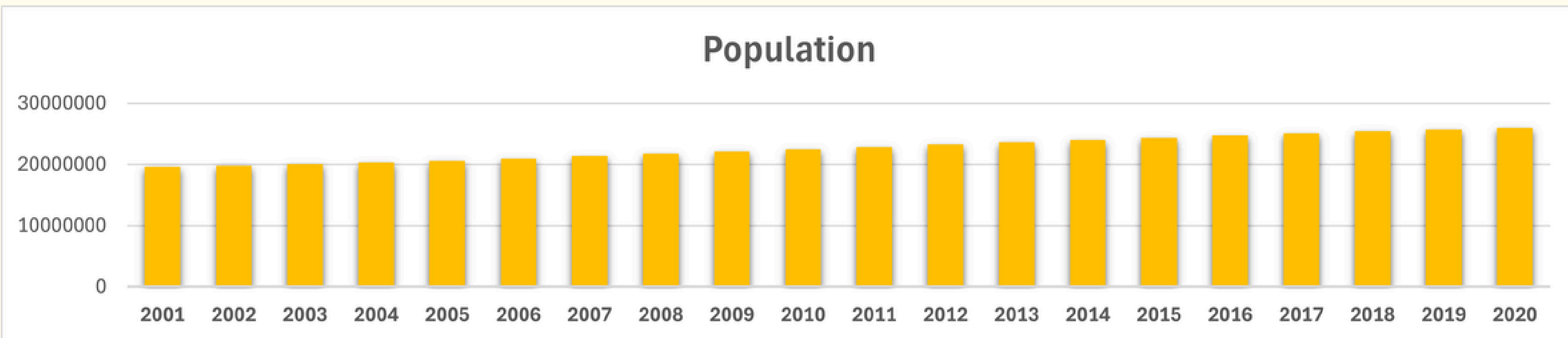
- Effect of changes in helathcare expenditure on population?
- What diseases are the leading causes of death in Australia?
- What diseases have the largest average cause of death?
- Whether there is a relationship between Health Expenditure and Disease Rate in Australian?
- Are there any irregularities that should be of particular concern from the results of this analysis?

Australia's

Healthcare expenditure for health care systems and social health insurance





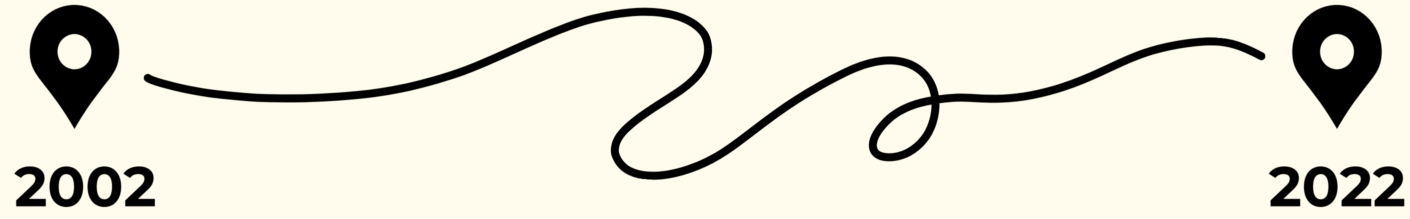


Source:

- Our World In Data based on Lindert (1994), OECD (1993), OECD Stat – processed by Our World in Data
- UN, World Population Prospects (2024) – processed by Our World in Data

Australia's

average of death rate and causes of death in the last 20 years

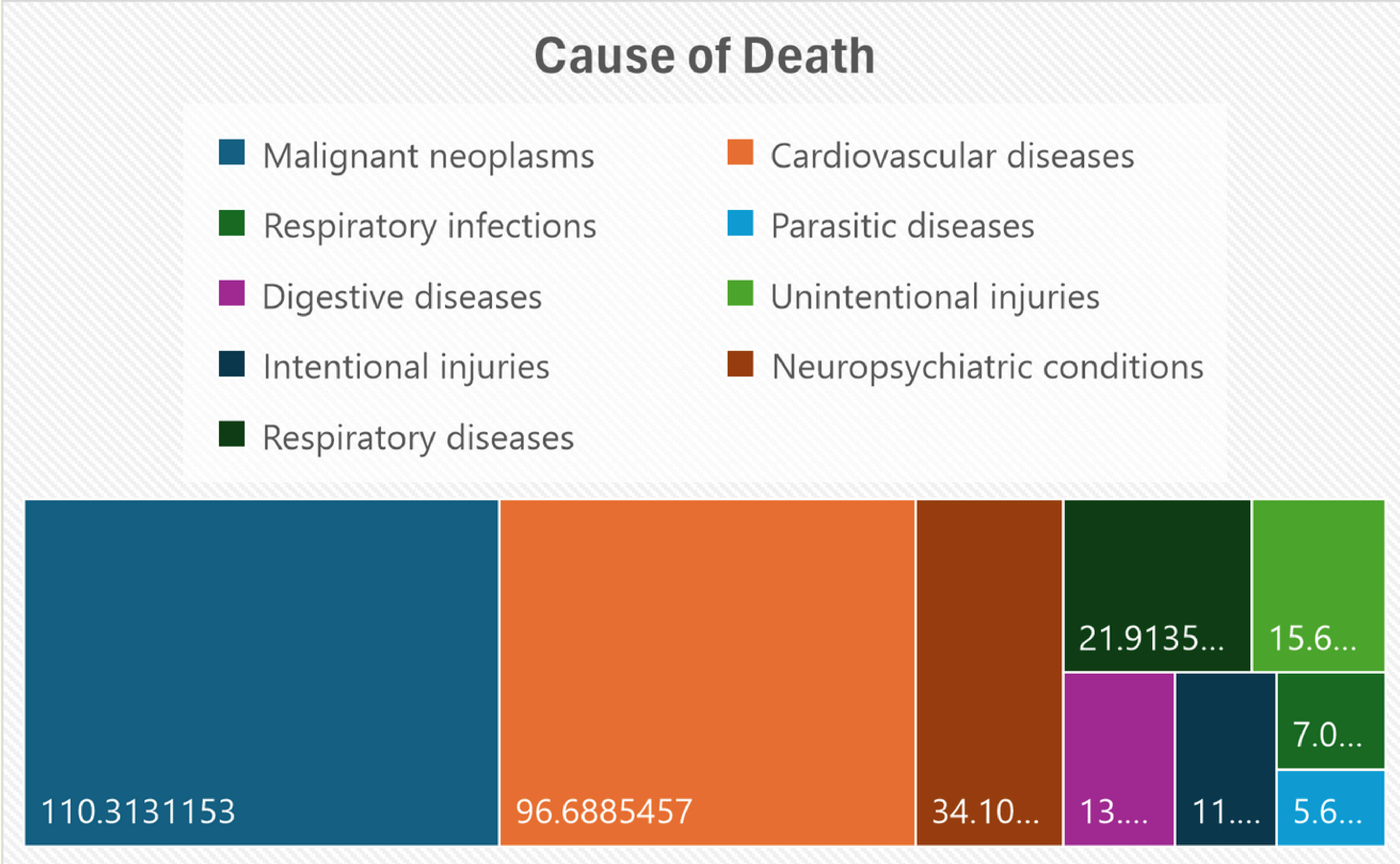


In many countries, when someone dies the cause of their death is determined by the medical practitioner who cared for them (such as a physician or nurse)

The World Health Organization (WHO) provides international guidelines called the International Classification of Diseases (ICD) to guide practitioners on how the cause of death should be described.

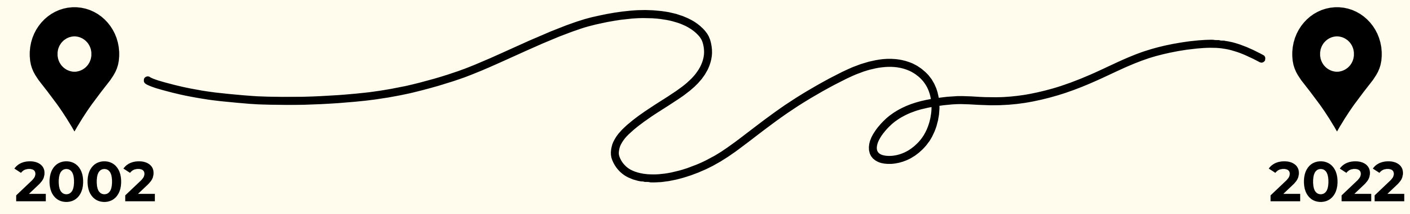
Source:

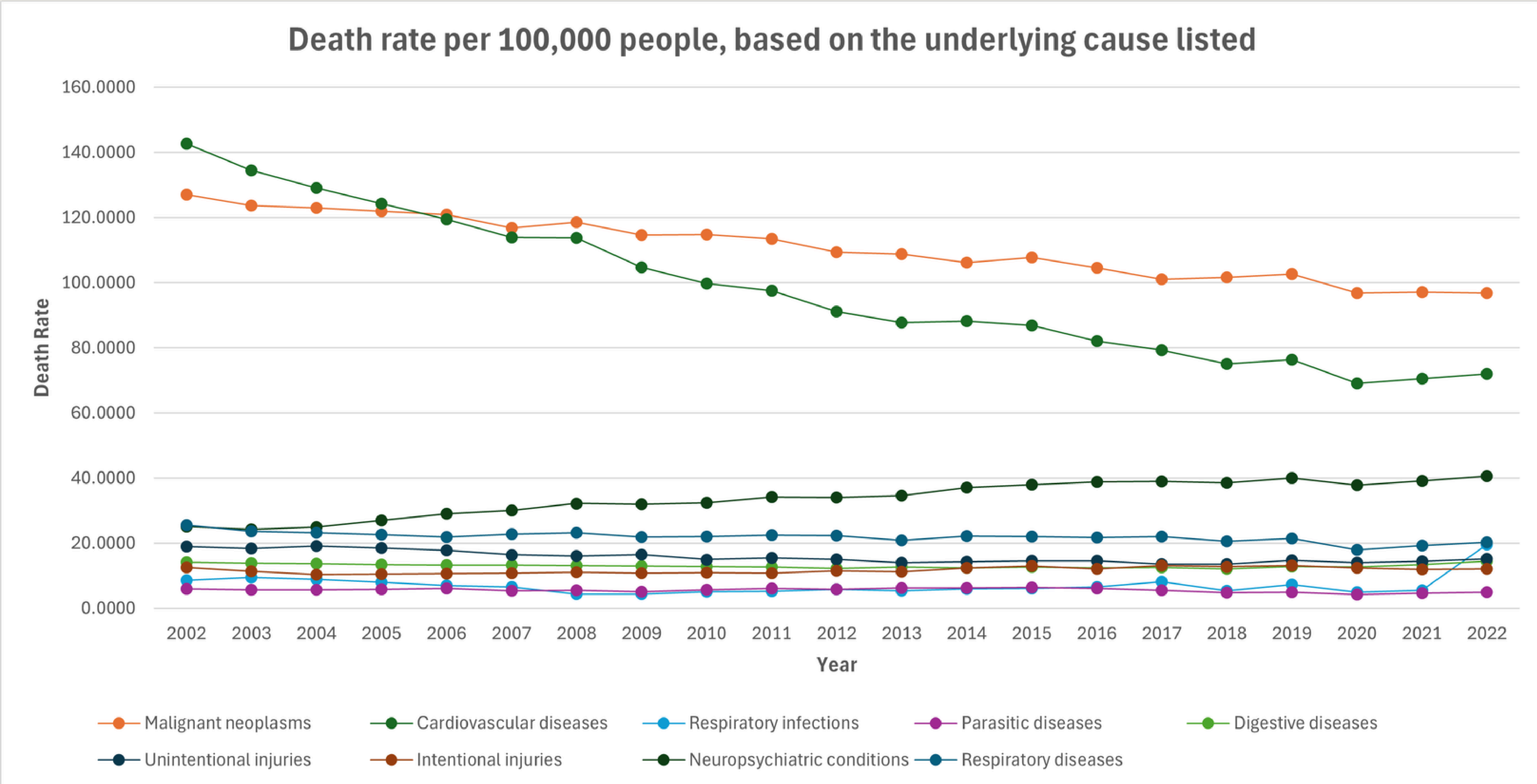
WHO Mortality Database (2024) – with minor processing by Our World in Data



Australia's

death rate and causes of death in the last 20 years





Regression Analysis

Between Healthcare Expenditure and Malignant Neoplasms

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.938896							
R Square	0.881525							
Adjusted R Square	0.87529							
Standard Error	3.388844							
Observations	21							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	1	1623.553	1623.553	141.3719	3.03E-10			
Residual	19	218.2011	11.48427					
Total	20	1841.754						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	188.138	6.540732	28.76406	3.96E-17	174.4481	201.8279	174.4481	201.8279
X Variable 1	-8.61205	0.724311	-11.89	3.03E-10	-10.128	-7.09605	-10.128	-7.09605

Regression Analysis

Between Healthcare Expenditure and Cardiovascular Diseases

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.902336							
R Square	0.81421							
Adjusted R Square	0.804431							
Standard Error	9.994959							
Observations	21							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	1	8318.183	8318.183	83.26576	2.25E-08			
Residual	19	1898.085	99.89921					
Total	20	10216.27						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	272.9027	19.29104	14.1466	1.54E-11	232.5261	313.2794	232.5261	313.2794
X Variable 1	-19.4934	2.136261	-9.12501	2.25E-08	-23.9646	-15.0222	-23.9646	-15.0222